

BASAVARAJESWARI GROUP OF INSTITUTIONS

BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT



NACC Accredited Institution*
(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to
Visvesvaraya Technological University, Belagavi)
"JnanaGangotri" Campus, No.873/2, Ballari-Hospet Road,
Allipur, Ballar1-583 104 (Karnataka) (India)
Ph: 08392 – 237100 / 237190, Fax: 08392 – 237197



DEPARTMENT OF CSE(DATA SCIENCE)

A Mini-Project Report On

**“TELECOM CUSTOMER CHURN AND REVENUE RISK
ANALYSIS USING POWER BI”**

A report submitted in partial fulfillment of the requirements for the

DATA VISUALIZATION WITH POWER BI

Submitted By

PRIYANKA G

USN: 3BR23CD071

**Under the Guidance of
Mr. Sundeep Sindhanoor**

Asst. Professor

**Dept of CSE (DATA SCIENCE),
BITM, Ballari**



Visvesvaraya Technological University

Belagavi, Karnataka 2025-2026

BASAVARAJESWARI GROUP OF INSTITUTIONS

BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

NACC Accredited Institution*

(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to
Visvesvaraya Technological University, Belagavi)

"JnanaGangotri" Campus, No.873/2, Ballari-Hospet Road, Allipur,
Ballari-583 104 (Karnataka) (India)

Ph: 08392 – 237100 / 237190, Fax: 08392 – 237197



DEPARTMENT OF CSE (DATA SCIENCE)

CERTIFICATE

This is to certify that the OPEN-ENDED PROJECT of DATA VISUALIZATION WITH POWER BI **“TELECOM CUSTOMER CHURN AND REVENUE RISK ANALYSIS USING POWER BI”** has been successfully presented by **PRIYANKA G** bearing USN **3BR23CD071** students of semester B.E for the partial fulfillment of the requirements for the award of **Bachelor Degree in CSE(DS)** of the BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT, BALLARI during the academic year 2025-2026.

Signature of guide

Mr.Sundeep Sindhanoor

Signature of HOD

Dr. Aradhana D

ACKNOWLEDGEMENT

The satisfactions that accompany the successful completion of our mini project on “

TELECOM CUSTOMER CHURN AND REVENUE RISK ANALYSIS USING POWER BI”

would be incomplete without the mention of people who made it possible, whose noble gesture, affection, guidance, encouragement and support crowned my efforts with success. It is our privilege to express our gratitude and respect to all those who inspired us in the completion of our mini- project.

We are extremely grateful to our Guides **Mr.Sundeep Sindhanoor** for their noble gesture, support co-ordination and valuable suggestions given in completing the mini-project. We also thank **Dr. Aradhana D**, H.O.D. Department of CSE(DS), for his co-ordination and valuable suggestions given in completing the mini-project. We also thank Principal, Management and non-teaching staff for their co-ordination and valuable suggestions given to us in completing the Open ended project.

Name

PRIYANKA G

USN

3BR23CD071

TABLE OF CONTENTS

Ch No	Chapter Name	Page
1	Introduction 1.1 About the project 1.2 Language used	1
2	Scope of the project 2.1 Problem statement 2.2 Objectives	2
3	System requirements 3.1 Functional Requirements 3.2 Non-Functional Requirements 3.3 Hardware Requirements 3.4 Software Requirements	3
4	Methodology 4.1 Data Collection 4.2 Data Preparation(Power Query) 4.3 Data Visualization 4.4 Filters and Interactivity 4.5 Dashboard Design	4-5
5	Implementation	6-9
6	Result	10-13
7	Conclusion References	13 14

ABSTRACT

This project focuses on analyzing customer churn in the telecom industry using data visualization techniques in Power BI. Customer churn is a critical issue faced by telecom companies, as losing customers directly impacts revenue and business growth. The objective of this project is to identify high-risk customers, analyze factors contributing to churn, and evaluate the potential revenue loss associated with it.

The dataset used in this project includes customer details such as contract type, tenure, internet service, monthly charges, and churn status. Using Power BI, various interactive dashboards and visualizations were created to study patterns and trends in customer behavior. Key performance indicators such as total customers, total churned customers, churn rate, and revenue at risk were calculated using DAX functions.

The analysis reveals that customers with month-to-month contracts, low tenure, and fiber optic internet services exhibit higher churn rates. Additionally, high-risk customers contribute significantly to revenue loss. The dashboard also highlights competitor switching behavior, providing insights into market competition.

This project demonstrates how data visualization can be effectively used to derive actionable insights and support decision-making. The findings can help telecom companies design better customer retention strategies and reduce churn. Future enhancements may include integrating machine learning models for predictive churn analysis.

CHAPTER 1

INTRODUCTION

1.1 About the Project

The Telecom Customer Churn and Revenue Risk Analysis Dashboard is designed to analyze customer churn behavior and identify revenue risk in the telecom industry using Microsoft Power BI Desktop. Customer churn directly affects business growth and profitability, making customer retention an important challenge for telecom providers.

This project helps identify customers who are most likely to leave the service and the major reasons behind churn. The dashboard analyzes important factors such as contract type, internet service type, tenure group, monthly charges, and customer switching behavior.

The analysis shows that customers with month-to-month contracts and fiber optic internet services have the highest churn rate. High-risk customers contribute the most to revenue loss, and most churned customers are switching to Jio, making it the top competitor.

The dashboard provides clear business insights through KPI cards, bar charts, pie charts, and competitor analysis. It helps management take better decisions for customer retention, reducing churn, and protecting revenue.

1.2 Language Used

The project is developed using:

- **DAX (Data Analysis Expressions)** – for creating calculated columns, measures, and KPIs such as Churn Rate, Revenue at Risk, and Top Competitor.
- **Power Query M Language** – for data cleaning, transformation, and creating custom columns in Power BI.
- **SQL concepts** – for understanding data relationships and structured data analysis.
- **Visualization tools in Microsoft Power BI Desktop** – for creating interactive dashboards and reports.

CHAPTER 2

SCOPE OF THE PROJECT

2.1 Problem Statement

Telecom companies face major financial losses due to customer churn, where customers discontinue services and switch to competitors. High churn rates reduce revenue and increase customer acquisition costs. Identifying high-risk customers and understanding the reasons behind churn is difficult using traditional reports. Therefore, there is a need for an interactive dashboard that can analyze churn patterns, calculate revenue at risk, and help management make better customer retention decisions.

2.2 Objectives

- To analyze customer churn patterns in the telecom industry using Microsoft Power BI Desktop.
- To identify high-risk customers who are likely to leave the service.
- To calculate revenue loss caused by customer churn.
- To analyze churn based on contract type, internet service, and tenure group.
- To identify the top competitor attracting churned customers.
- To provide business insights that help improve customer retention and reduce revenue loss.

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 Functional Requirements

- The system should display customer churn details such as total customers, churned customers, and churn rate.
- The dashboard should analyze churn based on contract type, internet service, and tenure group.
- The system should identify high-risk customers and calculate revenue at risk.
- The system should provide interactive filters and slicers for dynamic analysis
- The dashboard should show competitor analysis and the top competitor attracting churned customers.

3.2 Non-Functional Requirements

-
- The dashboard should be user-friendly and easy to understand.
- The system should provide fast and accurate visualization of data.
- The report should be visually clear and professionally designed
- The dashboard should maintain data consistency and reliability.
- The project should be easy to update and maintain in the future.

3.3 Hardware Requirements

- Processor: Intel Core i3 / i5 or higher
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: Minimum 20 GB free disk space
- System Type: 64-bit Windows operating system

3.4 Software Requirements

- Microsoft Power BI Desktop
- Microsoft Excel or CSV dataset file
- Windows 10 / Windows 11 Operating System

CHAPTER 4

METHODOLOGY

The telecom customer dataset was imported into Microsoft Power BI Desktop for analysis. Data cleaning and transformation were done using Power Query by handling missing values and creating new columns like Risk Level, Revenue at Risk, Tenure Group, and Switched To.

DAX measures were used to calculate KPIs such as Total Customers, Total Churned Customers, Churn Rate, and Revenue at Risk. Finally, interactive visuals like KPI cards, bar charts, pie charts, and slicers were created to analyze customer churn and support better business decisions.

4.1 Data Collection

The dataset used in this project contains telecom customer-related information such as:

- Customer ID
- Gender
- Contract Type
- Internet Service Type
- Monthly Charges
- Total Charges
- Tenure Group
- Churn Status
- Revenue at Risk
- Risk Level
- Competitor Switched To

4.2 Data Preparation (Power Query)

After loading the dataset into Power BI, data cleaning and transformation are performed using Power Query:

- Removed missing or null values
- Corrected data types for numerical and categorical fields

CHAPTER 4

- Created new columns such as Risk Level, Revenue at Risk, Tenure Group.

4.3 Data Visualization

The cleaned data is then used to create interactive visualizations in Microsoft Power BI Desktop. As shown in the dashboard:

Visuals Used:

- KPI Cards → Display total customers, total churned customers, churn rate, and revenue at risk
- Bar Chart → Shows churn analysis based on contract type and risk level
- Column Chart → Represents high-risk churn by tenure group and average monthly charges by internet type
- Pie Chart → Displays customer switching distribution by competitor
- Card Visual → Shows the top competitor attracting churned customers
- Table/Insight Box → Displays key business insights and summary analysis

4.4 Filters and Interactivity

Slicers and filters are added to make the dashboard interactive. Users can filter data based on internet service type, contract type, and tenure group to analyze specific customer segments and their churn patterns.

4.5 Dashboard Design

The final dashboard is designed to present all insights in a structured layout:

- Top section → KPI cards showing total customers, total churned customers, churn rate, and revenue at risk
- Middle section → Bar charts and column charts for contract-wise churn, risk level analysis, and tenure group analysis
- Right side → Pie chart and Top Competitor card showing customer switching behavior and competitor analysis
- Bottom section → Key insights and summary analysis for better business understanding

This layout improves usability and makes the dashboard easy to understand.

4.6 Outcome

The final dashboard helps identify high-risk customers, major churn reasons, revenue loss, and top competitors. It supports telecom companies in improving customer retention strategies and reducing financial losses.

CHAPTER 5

IMPLEMENTATION

The implementation of this project begins with importing the telecom customer dataset into Microsoft Power BI Desktop from CSV format. The dataset includes details such as customer ID, contract type, internet service type, monthly charges, total charges, tenure group, churn status, and revenue information. Data cleaning and transformation are performed using Power Query Editor by removing missing values, correcting data types, and creating new columns like Risk Level, Revenue at Risk, Tenure Group, and Switched To for competitor analysis.

After data preparation, DAX measures are created to calculate important KPIs such as Total Customers, Total Churned Customers, Churn Rate, Revenue at Risk, and Top Competitor. Interactive visuals like KPI cards, bar charts, pie charts, column charts, and slicers are then designed to analyze customer churn patterns, identify high-risk customers, and track competitor switching behavior. This helps telecom companies make better decisions for customer retention and revenue protection.

5.1 Dataset Description

The dataset used in this project consists of telecom customer-related information collected from customer service records. It includes the following fields:

- Customer ID – Unique identification number of each customer
- Gender – Gender of the customer
- Contract Type – Type of contract such as Month-to-Month, One Year, or Two Year
- Internet Service Type – Type of internet service such as DSL, Fiber Optic, Cable, or No Internet
- Monthly Charges – Monthly bill amount of the customer
- Total Charges – Total amount paid by the customer
- Tenure Group – Duration of customer service usage
- Churn Status – Indicates whether the customer has left the service or not
- Risk Level – Customer risk category such as High, Medium, or Low

CHAPTER 5

- Revenue at Risk – Estimated revenue loss due to churn
- Switched To – Competitor chosen by churned customers

5.2 Data Loading

The dataset is loaded into Microsoft Power BI Desktop using the “Get Data” option.

- Data is imported from Excel/CSV format
- Customer records are previewed and verified before processing
- Required tables and fields are selected for analysis

5.3 Data Transformation (Power Query)

Using Power Query Editor, the dataset is cleaned and prepared:

- Removed null and duplicate values
- Renamed columns for better clarity
- Corrected data types for text, numeric, and categorical fields
- Created new columns such as Risk Level, Revenue at Risk, Tenure Group, and Switched To
- Ensured consistency across contract type, internet service type, and churn status

5.4 Data Modeling

- No complex relationships or ER diagram are required
- All customer data fields are directly used for visualization
- Calculated columns and DAX measures are created within the same table
- This approach keeps the dashboard simple, efficient, and easy to manage

5.5 DAX Measures (Key Calculations)

Total Customers = COUNT('final_churn_dataset'[Customer ID])

Total Churned Customers =
CALCULATE(COUNT('final_churn_dataset'[Customer ID]),
'final_churn_dataset'[Churn] = "Yes")

Churn Rate =
DIVIDE([Total Churned Customers], [Total Customers], 0) * 100

Revenue at Risk =
SUM('final_churn_dataset'[Revenue_at_Risk])

CHAPTER 5

These measures are used in KPI cards, charts, and competitor analysis visuals.

5.6 Dashboard Visualizations

The dashboard includes multiple visuals:

Main Visuals:

- **KPI Cards**

Display total customers, total churned customers, churn rate, and revenue at risk.

- **Bar Chart (Churn by Contract Type)**

Helps compare customer churn across different contract types such as Month-to-Month, One Year, and Two Year.

- **Column Chart (High-Risk Churn by Tenure Group)**

Shows churn distribution across different customer tenure groups.

- **Pie Chart (Customer Switching Distribution)**

Represents the percentage of churned customers switching to competitors like Jio, Airtel, Vodafone, and BSNL.

- **Card Visual (Top Competitor)**

Displays the top competitor attracting the highest number of churned customers.

5.7 Filters /Slicers

To enhance interactivity, the following filters are added:

- Internet Service Type
- Contract Type
- Tenure Group
- Churn Status

Users can dynamically filter data and analyze specific customer segments, churn patterns, and competitor switching behavior.

5.8 Dashboard Layout

The dashboard is designed in a structured way:

- Top section → KPI cards showing Total Customers, Total Churned Customers, Churn Rate, and Revenue at Risk
- Middle section → Bar charts and column charts for contract-wise churn, risk level analysis, and tenure group analysis
- Right side → Pie chart and Top Competitor card for customer switching distribution and competitor analysis
- Bottom section → Summary insights and key business findings

This layout ensures better readability, smooth navigation, and improved user experience.

5.9 Integration

Data is imported from Excel/CSV format

All visuals are connected to the same telecom customer dataset

DAX measures and calculated columns are linked for KPI analysis

Filters and slicers dynamically update all visuals for interactive analysis

5.10 Final Output

The implementation results in a fully interactive Microsoft Power BI Desktop dashboard that:

- Displays customer churn patterns and revenue risk analysis
- Allows contract-wise, internet service-wise, and tenure-based analysis
- Helps identify high-risk customers and revenue loss segments
- Provides clear visual insights on competitor switching behavior and top competitor analysis

CHAPTER 6

RESULT

The Telecom Customer Churn and Revenue Risk Analysis Dashboard successfully identifies major customer churn patterns and revenue loss areas in the telecom industry. The analysis shows that out of 7042 total customers, 1869 customers have churned, resulting in a churn rate of 26.54%. The total revenue at risk is 139.13K, indicating a significant financial impact on the company.

The dashboard reveals that customers with month-to-month contracts and fiber optic internet services have the highest churn rate. High-risk customers contribute the most to revenue loss, especially those in the 0–12 months tenure group. Competitor analysis shows that most churned customers are switching to Jio, making it the top competitor. These insights help telecom companies improve customer retention strategies and reduce revenue loss effectively.

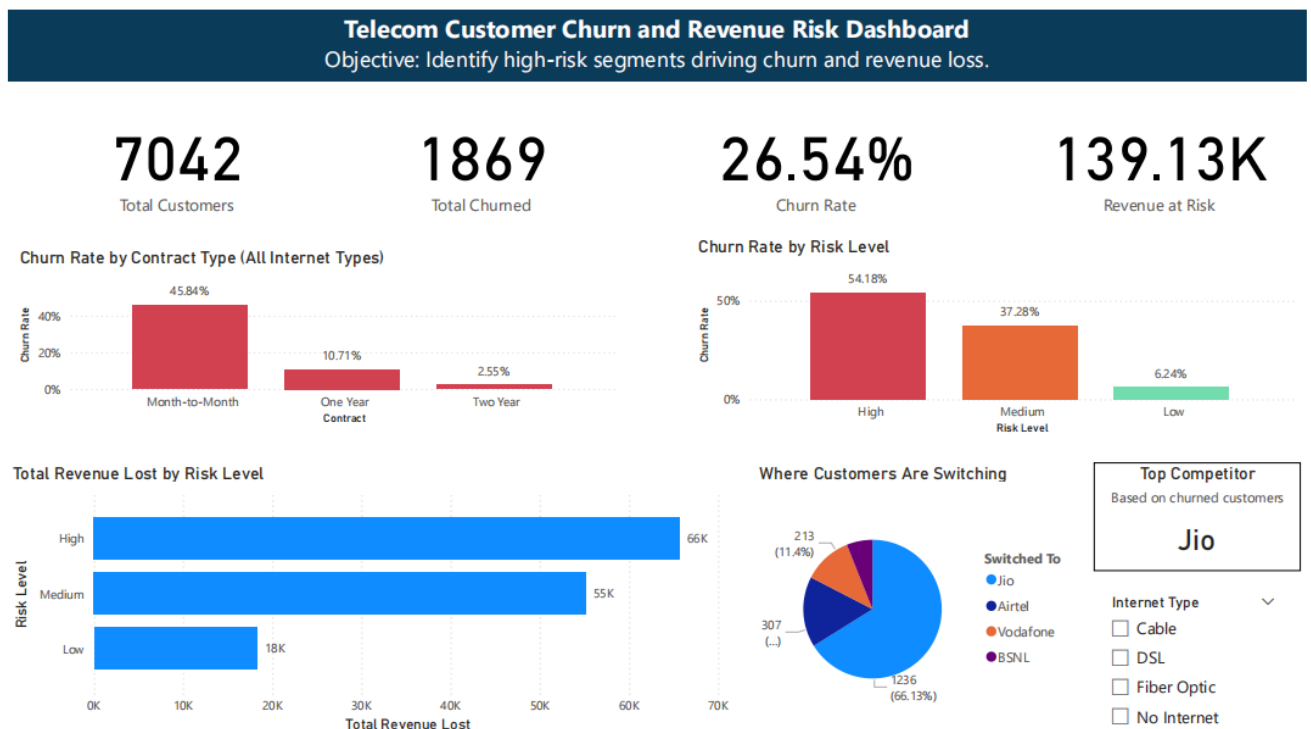


Figure 1: Telecom Customer Churn and Revenue Risk Dashboard

CHAPTER 6

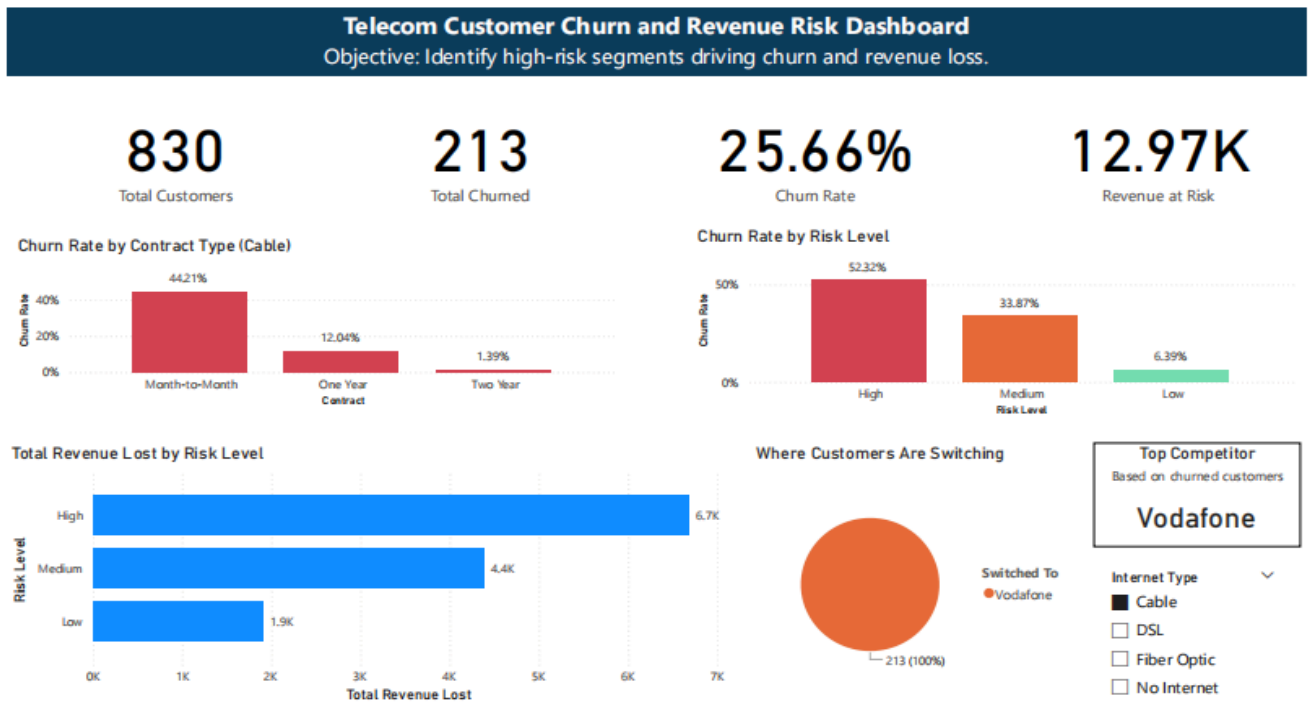


Figure 2: Telecom Customer Churn and Revenue Risk Dashboard for Cable Internet Service

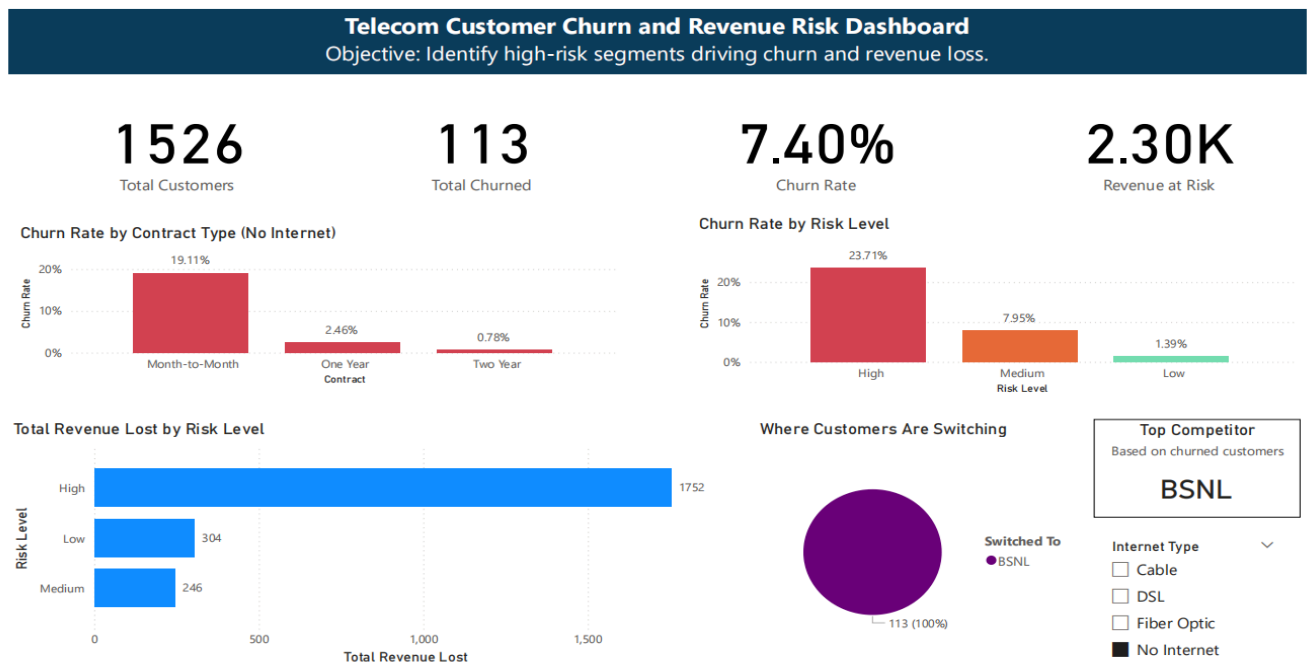


Figure 3: Telecom Customer Churn and Revenue Risk Dashboard for No Internet

CHAPTER 6

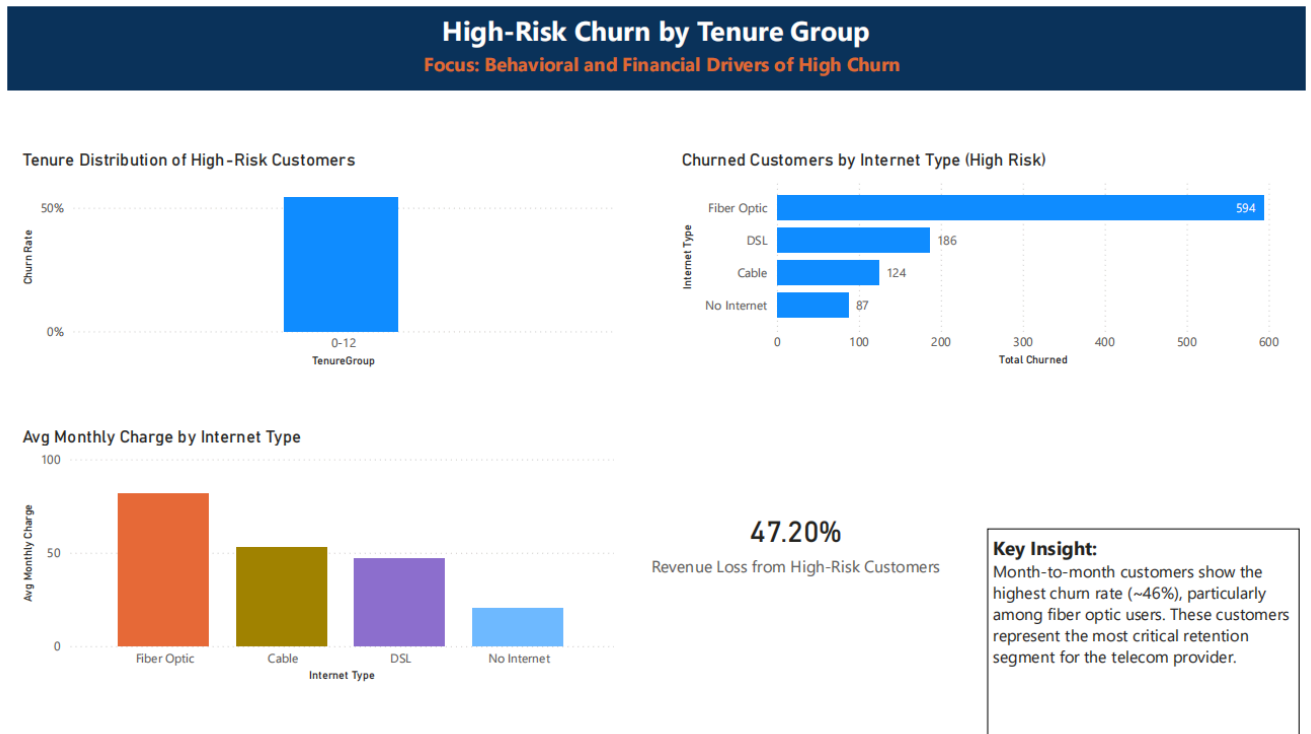


Figure 4: High-Risk Churn Analysis by Tenure Group Dashboard

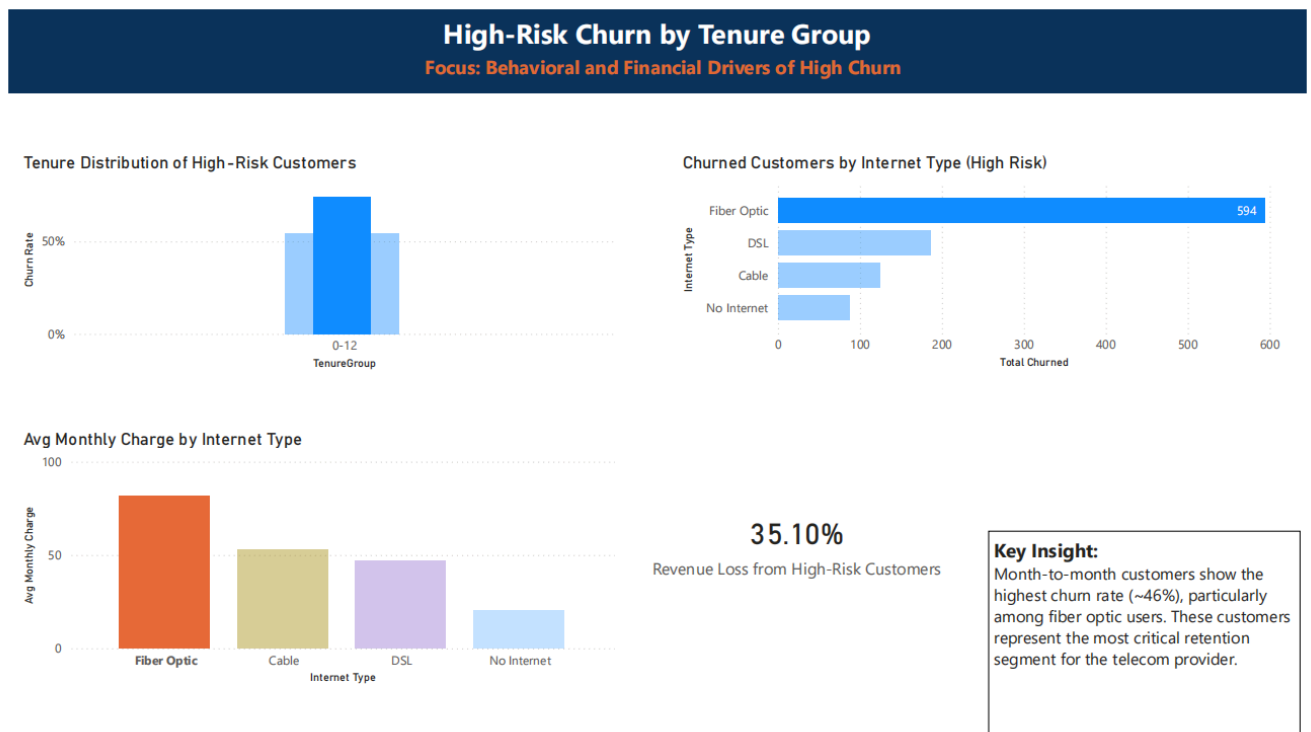


Figure 5: High-Risk Churn Analysis for Medium Risk Customers by Tenure Group

CHAPTER 7

CONCLUSION

The Telecom Customer Churn and Revenue Risk Analysis Dashboard provides an effective solution for understanding customer behavior, identifying churn patterns, and analyzing revenue risk in the telecom industry. By using Power BI, the project transforms raw customer data into meaningful insights through interactive dashboards, KPI cards, charts, and competitor analysis. This makes the analysis simple, visual, and easy for business users to understand.

The results show that customers using month-to-month contracts and fiber optic internet services are more likely to leave the service. Customers with shorter tenure periods, especially in the 0–12 months group, also show higher churn rates. High-risk customers contribute significantly to revenue loss, which directly affects business profitability. The dashboard also reveals that most churned customers are switching to Jio, making it the strongest competitor in the market.

This project helps telecom companies focus on high-risk customer segments, improve customer satisfaction, and design better retention strategies such as loyalty benefits, contract upgrades, and personalized offers. It also supports management in making faster and smarter business decisions based on real-time visual insights. Overall, the project improves operational efficiency and helps reduce customer churn while protecting company revenue.

REFERENCES

- [1] Microsoft, “Microsoft Power BI Documentation,” Microsoft Learn. [Online]. Available: <https://learn.microsoft.com/>.
- [2] Kaggle, “Telecom Customer Churn Dataset,” Kaggle. [Online]. Available: <https://www.kaggle.com/>.
- [3] DataFlair, “Power BI Tutorial for Beginners,” DataFlair. [Online]. Available: <https://data-flair.training/>. [Accessed: Apr. 2026]
- [4] GeeksforGeeks, “Power BI and DAX Functions,” GeeksforGeeks. [Online]. Available: <https://www.geeksforgeeks.org/>.
- [5] Tutorials Point, “Business Intelligence and Power BI Concepts,” Tutorials Point. [Online]. Available: <https://www.tutorialspoint.com/>.
- [6] Microsoft, “DAX Functions Reference,” Microsoft Learn. [Online].
- [7] Various Authors, “Research Papers on Telecom Customer Churn Analysis and Customer Retention Strategies,” Google Scholar. [Online]. Available: <https://scholar.google.com/>.